

CoRoT-2b

R-1657

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_130927 · created 2026-08-02T08:52:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456562.6947 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.222 +/- 0.062
Transit depth (Rp/Rs)^2	4.91 +/- 2.74 [%]
Orbital Inclination (inc)	84.17 +/- 1.89
Airmass coefficient 1 (a1)	1.0 +/- 0.14
Airmass coefficient 2 (a2)	0.028 +/- 0.038
Scatter in the residuals of the lightcurve fit is	14.45 %
Best Comparison Star	#3 - [163, 211]
Optimal Aperture	2.87
Optimal Annulus	12.89
Transit Duration (day)	0.082 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.222 ± 0.062 vs 0.1667 (deviation 0.73σ)
Duration fit vs published	0.082 d vs 0.095 d (deviation 0.66σ)
Mid-time O–C	0.3 min
Depth SNR	2.9
Residual scatter	14.45 %

Light curve

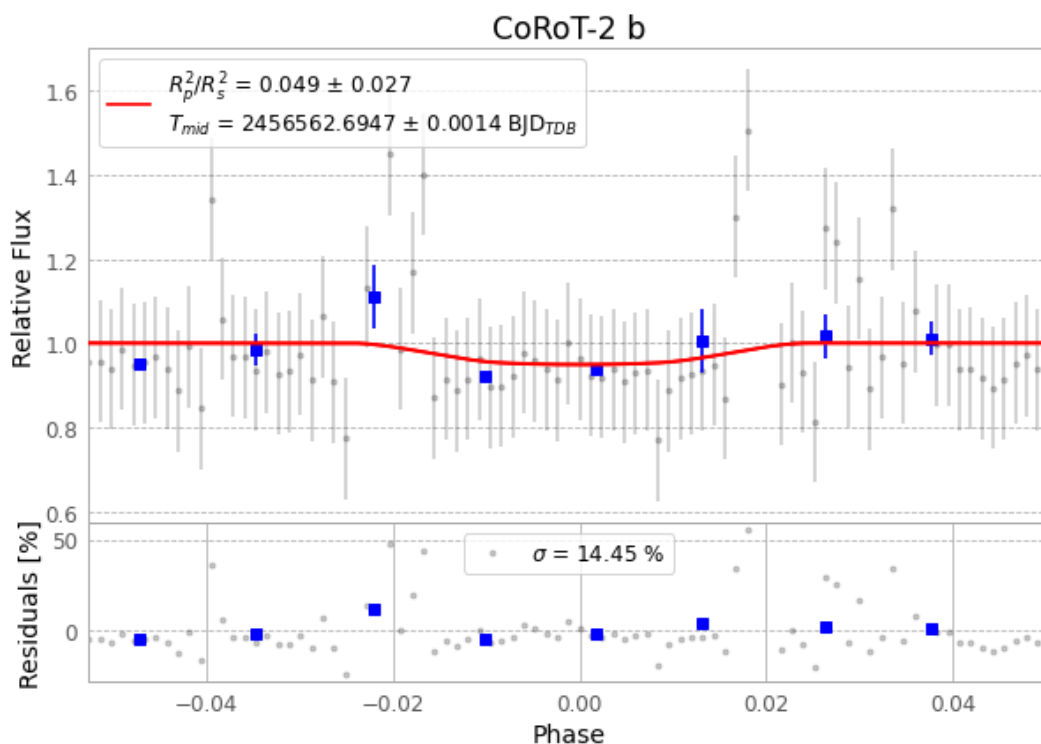


Figure 1 — FinalLightCurve_CoRoT-2 b_2013-09-26.png (SHA-256 106cf96c23ed7f4f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d35e2d1c4e38a7ea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), CoRoT-2130927022412.FITS → CoRoT-2130927064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-130927.log (ed1fe173e790...), FinalLightCurve_CoRoT-2 b_2013-09-26.png (106cf96c23ed...), FinalLightCurve_CoRoT-2 b_2013-09-26.csv (e38fbb9f852f...)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 08:52Z.